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ORIGINAL RESEARCH



Age differences in outcome after mild traumatic brain injury: results from the HeadSMART study

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ABSTRACT

This study longitudinally examined age differences across multiple outcome domains in individuals diagnosed with acute mild traumatic brain injury (mTBI). A sample of 447 adults meeting VA/DoD criteria for mTBI was dichotomized by age into older (≥ 65 years; $n = 88$) and younger (< 65 years; $n = 359$) sub-groups. All participants presented to the emergency department within 24 hours of sustaining a head injury, and outcomes were assessed at 1-, 3-, and 6-month intervals. Depressive symptoms were assessed using the Patient Health Questionnaire-9 (PHQ-9), post-concussive symptoms (PCS) were ascertained with the Rivermead Post-Concussion Questionnaire (RPQ), and functional recovery from the Extended Glasgow Outcome Scale (GOSE). Mixed effects logistic regression models showed that the rate of change over time in odds of functional improvement and symptom alleviation did not significantly differ between age groups ($p = 0.200$ – 0.088). Contrary to expectation, older adults showed equivalent outcome trajectories to younger persons across time. This is a compelling finding when viewed in light of the majority opinion that older adults are at risk for significantly worse outcomes. Future work is needed to identify the protective factors inherent to sub-groups of older individuals such as this.

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Mild traumatic brain injury; mTBI; age differences; older adults; outcomes research

Introduction

Diverse areas of medicine are now claiming individual variability as the cornerstone of novel approaches to treatment and prevention. This recent emphasis is contrary to prior views that sought to diminish personal differences as excess noise in data-driven ventures. The traumatic brain injury (TBI) field has also begun delving into precision medicine approaches involving genetics, biomarkers, and specific mechanisms of injury. Despite this, there is startlingly little work dedicated to the impact of geriatric-related factors on TBI outcomes. The attention specified towards older adult TBI populations pales in comparison to the massive presence of sport- and military-related TBI.

Older adults have the highest rate of TBI-related deaths and number of hospitalizations compared to

all other age groups, and the largest number of emergency department visits (tied with that of the paediatric population) (Taylor, Bell, Breiding, & Xu, 2017). Emergency medical care, death, or hospitalization resulting from TBI will impact one in 200 American adults aged 65–74 and one in 50 aged 75 years and above (Taylor et al., 2017). There is also evidence that these phenomena represent an increasing trend rather than a static fact. In 2009 and 2010, ~39 million older adults were evaluated for a newly-obtained TBI in US emergency departments, which represented a 61% increase from prior estimates (Albrecht et al., 2016). Increased age is also associated with higher rates of anticoagulant use, polypharmacy, anatomical frailty, comorbid conditions including neurodegenerative disorders, and falls as a mechanism of injury, which calls for a focused approach to TBI

management (Peters & Gardner, 2018). Given the compelling epidemiological case for geriatric TBI, further research in this area is warranted, especially when focusing on factors that may differ between older and younger patients.

The majority of current literature asserts a linear association between severity of brain injury-related outcomes and increased age (Czosnyka et al., 2005; Hukkelhoven et al., 2003; LeBlanc, de Guise, Gosselin, & Feyz, 2006; Livingston et al., 2005; Mushkudiani et al., 2007; Perdue, Watts, Kaufmann, & Trask, 1998) indicating that older adults with TBI may generally be an at-risk population. In terms of outcome, the culmination of TBI treatment efforts is the patient's overall functional ability; that is, how well a person can actively engage in basic and complex life activities in comparison to their unique pre-injury baseline status. Functional outcome seems to be a particularly potent factor in the geriatric TBI equation, as pre-admission functional ability has high prognostic significance for post-injury functional recovery and rehabilitation efficiency (Yu & Richmond, 2005). There is also evidence that older adult TBI survivors have greater disability-associated dependence than younger TBI groups; a finding that has been validated across multiple global outcome measures (Thompson, McCormick, & Kagan, 2006). The majority of elderly persons do eventually attain complete functional recovery, though this process may take from 1–3 years (Abdulle et al., 2018). There is evidence that the pace of functional change demonstrated by aged populations is at a significantly slower rate than that of younger groups (Frankel et al., 2006).

The monocular viewpoint concerning older adult TBI is not surprising given that higher age is associated with increased risk and mortality in a variety of neuropsychiatric disorders (Witlox et al., 2010), forms of trauma other than TBI (Perdue et al., 1998), and medical prediction algorithms involving chronic disease (Charlson, Szatrowski, Peterson, & Gold, 1994). However, assuming that this trend applies to all clinical situations involving ageing populations is a potentially premature conclusion. Especially when considering the current medical landscape's emphasis on precision medicine, generalizations such as this should be met with an appropriate degree of reticence.

Contrary to the predominant perspective, an emerging set of evidence suggests that increased age may not have such a profound negative effect on TBI outcomes as originally thought, and that it may actually contribute protective factors. Results generated

from the authors' previous paper (Peters et al., 2018) found that older adults had better functional outcomes, fewer post-concussive symptoms (PCS), and less depressive symptomatology at 1 month post-injury in comparison to younger individuals. These findings resulted from a sample with confirmed blunt trauma of any severity who met American College of Emergency Physicians guidelines for a head CT. The results persisted after narrowing the sample to those who were clinically diagnosed with any severity of TBI using VA/DoD criteria, and in the case of functional recovery and PCS, older adults unexpectedly showed a superior outcome to younger cohorts. These findings revealed that the study population definition may be one source of the variability found in the literature on the effect of age on TBI outcomes.

The authors' prior work does not stand alone in the literature. In line with our previous results, adults aged 60 plus had better scores on measures of global outcome and mood/physical-related post-concussive symptoms than younger adults within 1 month of sustaining a mild TBI (mTBI) (Rapoport & Feinstein, 2001). Another study found older adults to have lower rates of depression than younger individuals in the acute phase following mTBI (Rapoport, McCullagh, Streiner, & Feinstein, 2003). Additionally, neurocognitive PCS in the sub-acute phase of geriatric mTBI may not be exclusively worse in comparison to younger groups (Stapert, Houx, De Kruijk, Ponds, & Jolles, 2006). There is a notable level of insight that can be gained from these studies; a sub-population of older adults with heightened resiliency in the short-term following a TBI may exist.

Given the previously referenced, biased state of the ageing literature (i.e. an inverse association between age and TBI outcomes), it is difficult to surmise the mechanism behind these contrary findings. One intriguing biologically-driven explanation is that age-related brain changes to anatomy or metabolism that are detrimental to behavioural and eloquent functions might be counterintuitively protective in other contexts. For example, cortical atrophy could indirectly soften the blow of a blunt head impact due to increased cerebrospinal fluid volume. While many investigators are intently interested in brain atrophy as a product of or having an altered course due to TBI, none that we are aware of have proposed the opposite—the impact of pre-existing atrophy upon TBI sequelae. Alternatively, very few outcome measures are validated for use in older adult TBI, which may inhibit outcomes from being accurately quantified. This phenomenon has already been observed in

Table 1. Baseline demographic comparison of older vs younger individuals with mild traumatic brain injury.

Characteristic ^a	Younger: Age <65 (n = 359)	Older: Age ≥65 (n = 88)	Test statistic ^b	p-value
Age, median in years (IQR)	37.2 (13.6)	75.7 (7.9)	-25.43	<0.001
Sex			17.78	<0.001
Male	216 (60.2)	31 (35.2)		
Female	143 (39.8)	57 (64.8)		
Race			46.90	<0.001
Black	175 (48.7)	13 (14.8)		
White	156 (43.5)	74 (84.1)		
Other	28 (7.8)	1 (1.1)		
Highest level of education			5.34	0.067
Less than High School	58 (16.2)	23 (26.1)		
High School graduate	223 (62.1)	45 (51.1)		
College graduate	78 (22.1)	20 (22.7)		
Married	93 (25.9)	34 (38.6)	5.63	0.024
Employed	216 (60.2)	11 (12.5)	64.25	<0.001
Prior concussion ^c	110 (30.6)	20 (22.7)	2.15	0.152
Mood disorder ^c	102 (28.4)	32 (36.4)	2.33	0.321
Non-mood psychiatric disorder ^c	80 (22.3)	14 (15.9)	2.01	0.392

^an (%) unless listed otherwise.^bFisher's exact for proportions and t-test for continuous variables.^cData were ascertained through self-report.

the commonly used Glasgow Coma Scale, which produces a dissimilar distribution for older vs younger adults. Elderly individuals consistently score higher (i.e., less impaired) despite having a larger burden of acute intracranial abnormality (Kehoe, Rennie, & Smith, 2015).

To gain a cleaner view of functional, mood, and post-concussive outcomes and in the interest of addressing limitations from our previous work, the current paper features a more specific, narrowed sample. By dichotomizing our sample into older (≥65 years) and younger (<65 years) sub-groups, this study aims to look at age differences in outcomes over 1-, 3-, and 6-month time points in individuals meeting Veterans Affairs and Department of Defense (VA/DoD) diagnostic criteria for mTBI (O'Neil et al., 2013). We hypothesize that, with time, younger people will have equivalent outcomes to older adults. That is, older adults at 1 month are likely to have less sequela than younger persons, but are expected to exhibit a delayed onset of symptoms as time progresses, ultimately resulting in incomplete recovery at 6 months. As a secondary objective, we assessed for differences in demographic and clinical characteristics between younger and older individuals meeting VA/DoD criteria for mTBI.

Materials and methods

Participants

Our sample is a sub-set of individuals from the Head Injury Serum Markers for Assessing Response to Trauma (HeadSMART) study. HeadSMART was a prospective cohort study recruiting from two Johns Hopkins emergency departments in Baltimore, MD

that examined the utility of blood-based biomarkers for diagnosing TBI and for early prediction of TBI outcome in adults (Peters et al., 2017). Recruitment for HeadSMART occurred between April 2014 and April 2017. All study activities were approved by and executed under the purview of a Johns Hopkins Institutional Review Board. Informed consent was gained from all participants.

Building upon our previous paper, this study applied several methodological alterations to its comparison of older vs younger individuals with head injury. First, all outcome domains were analysed over two additional time points, enabling comparison between 1-, 3-, and 6-months post-acute status. Second, those with moderate and severe head injury were excluded from the sample, leaving only those with mTBI. Classification of mTBI in elderly samples has shown differential prognostic significance from moderate TBI on a variety of neurobehavioural metrics (Goldstein, Levin, Goldman, Clark, & Altonen, 2001). The overall cohort was then dichotomized by age into younger (<65 years) and older (≥65 years) sub-samples. The age cut-off of 65 years was chosen due to its common occurrence in the ageing literature as well as being an American College of Emergency Physicians (ACEP) criterion for head CT after blunt head trauma (Jaoda, Bazarlan, Burns, & Al., 2008).

The current study's participants were narrowed to include only those who met VA/DoD criteria for mTBI (O'Neil et al., 2013). It is central to our design that all participants had not just a confirmed head injury, but a definitive clinical diagnosis of TBI based on stringent diagnostic criteria. Using this formalized severity system extends beyond simply denoting the occurrence of a concerning blow to the head, and

Table 2. Injury descriptor comparison of older vs younger individuals with mild traumatic brain injury.

Characteristic ^a	Younger: Age <65 (n = 359)	Older: Age ≥65 (n = 88)	Test statistic ^b	p-value
Mechanism of Injury			83.02	<0.001
Pedestrian Struck	44 (12.3)	2 (2.3)		
Motor Vehicle-Traffic	109 (30.4)	18 (20.5)		
Fall	84 (23.4)	64 (72.7)		
Assault	69 (19.2)	2 (2.3)		
Struck by/against	19 (5.3)	1 (1.1)		
Bicycle	32 (8.9)	1 (1.1)		
Other	2 (0.6)	0 (0.0)		
Intoxicated on Drugs/Alcohol	80 (22.3)	11 (12.5)	4.17	0.054
Symptoms at time of presentation				
GCS < 15 at presentation	44 (12.3)	8 (9.1)	0.95	0.569
Post-Traumatic Amnesia	190 (52.9)	37 (42.1)	4.03	0.134
Deficits in Short-term Memory ^c	30 (8.4)	8 (9.1)	1.23	0.430
Focal Neurological Deficits	23 (6.4)	3 (3.4)	1.42	0.554
Headache	306 (85.2)	64 (72.7)	7.76	0.007
Vomiting since injury	39 (10.9)	5 (5.7)	2.14	0.166
Head AIS score			3.65	0.205
Minor	340 (94.7)	79 (89.7)		
Moderate	9 (2.5)	3 (3.4)		
Serious	8 (2.2)	5 (5.7)		
Severe, Critical, or Maximal	2 (0.6)	1 (1.1)		

^an (%) unless listed otherwise.^bFisher's exact for proportions and *t*-test for continuous variables.^cData were ascertained in the acute period through self-report.

GCS: Glasgow Coma Scale; AIS: Abbreviated Injury Scale.

presumes that this blow has resulted in injury to the brain. VA/DoD criteria were chosen because, unlike some other classification systems, it integrates CT findings. Absence vs presence of acute abnormalities on CT can be an important way of partitioning mTBI from other injury severities. VA/DoD criteria for mTBI also include a Glasgow Coma Scale (GCS) score of 13–15 (best up to 24 h), loss of consciousness not exceeding 30 min, and alteration of conscious/mental status and post-traumatic amnesia duration of absent to 24 h.

Initial data collection

Collection of demographic (Table 1) and injury-related clinical information (Table 2) occurred at the enrolment visit in the emergency department based on recommendations provided by the National Institute of Neurological Disorders and Stroke (NINDS) common data elements for TBI (CDE v.2) (Hicks et al., 2013). Post-injury baseline data included the measurement of head injury severity (head Abbreviated Injury Scale [head AIS]) (Gennarelli & Wodzin, 2006), Glasgow Coma Scale score, neurological sequelae, mechanism of injury, and intoxication status at time of injury. All CT imaging studies were read by a single board-certified neuroradiologist.

Assessment of outcomes

Outcome data were collected 1-, 3-, and 6-months after enrolment via telephone interview or in-person

assessment. Outcome domains of interest were post-concussive symptoms (PCS), depressive symptomatology, and functional recovery defined as resumption of daily life activities. Dichotomized variables were created for each outcome. Information on PCS was ascertained using the Rivermead Post-Concussion Questionnaire (RPQ) (King, Crawford, Wenden, Moss, & Wade, 1995). The RPQ groups PCS into several factors (cognitive, emotional, somatic, other) and, in comparison to pre-morbid levels, rates the degree to which the symptom(s) have been problematic over the last 24 hours. By ICD-10 criteria, participants were deemed to have PCS if they reported at least mild, moderate, and/or severe problems in two or more of the following domains: (1) headaches, dizziness, general malaise, excessive fatigue, or noise intolerance; (2) irritability, emotional lability depression, or anxiety; (3) subjective complaints of concentration or memory difficulty; (4) insomnia; (5) reduced tolerance to alcohol; (6) pre-occupation with these symptoms and fear of permanent brain damage. Severity of depressive mood was measured by the Patient Health Questionnaire-9 (PHQ-9) (Kroenke, Spitzer, & Williams, 2001). The presence of at least mild depressive symptomatology was defined here as a score of ≥5, which has been found to produce adequate reliability and validity in detecting depression following TBI (Fann et al., 2005). The Extended Glasgow Outcome Scale (GOSE) (Jennett & Bond, 1975) ascertains an individual's level of functional recovery based on level of immersion in common everyday tasks such as working and

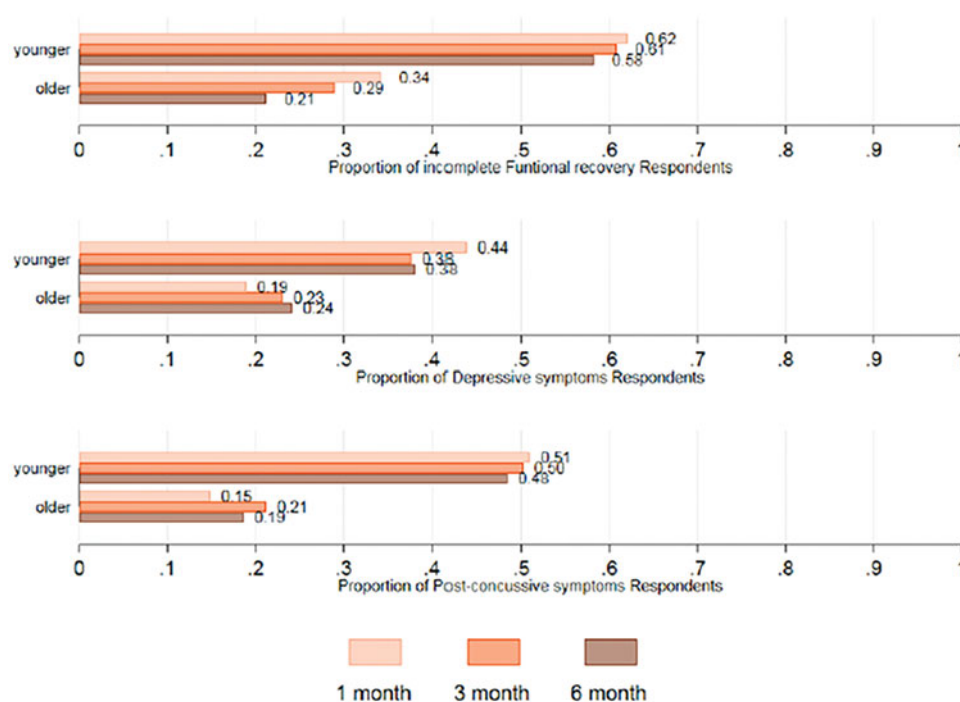


Figure 1. Bar chart of age effects on functional recovery, depressive symptoms, and post-concussive symptoms.

engaging with one's community and social network. Recovery is measured on a scale of 1 (deceased) to 8 (upper good recovery), with incomplete functional recovery defined as GOSE <8.

Analysis

Statistical analyses were performed in StataSE Version 13.0 (College Station, TX) (StataCorp, 2013). To assess for significance between groups differences on demographic and injury descriptors, *t*-tests were used for continuous variables and Fisher's Exact Test was used for categorical variables. Proportions were generated to compare the relative frequency of incomplete functional recovery, depression symptoms, and PCS across each age group and time point. Longitudinal mixed effects logistic regression models were then estimated to evaluate odds of each outcome over time as a function of age. The Missing at Random (MAR) assumption was used for missing data.

Results

Participant characteristics

Of the 554 participants with blunt head trauma enrolled in HeadSMART, 437 were <65 years of age (younger) and 117 were ≥65 years of age (older). Of those individuals, 359 younger (median age = 37.2)

and 88 older (median age = 75.5) participants met criteria for mTBI and were included in the present analyses. Detailed information on participant characteristics is displayed in Table 1. There were significant differences across age groups for marital status ($p=0.024$) as well as employment status, race, and sex ($p<0.001$). Endorsement of prior concussions or pre-existing psychiatric disorders did not significantly differ between age groups.

Mechanism of injury had a statistically significant difference between the two age groups; older adults were more likely to have fallen, while younger adults were most commonly affected by motor vehicle/traffic accidents. At time of presentation, participants did not differ substantially in GCS scores or intoxication status. Symptoms were generally not different between groups, with the only exception being that younger individuals reported significantly more headaches than older individuals. Headache and post-traumatic amnesia were the most common PCS endorsed across both age groups. Importantly, head injury severity score (head AIS) did not significantly differ as a function of age. See Table 2 for finer details regarding symptoms and mechanism of mTBI in our sample.

Logistic regression modelling of outcomes

Figure 1 shows rates of outcomes in older vs younger adults over time. Percentages reflect observed

Table 3. One-, 3-, and 6-month outcomes in older vs younger participants with mild traumatic brain injury.

Outcome	One month		Three months		Six months	
	Younger: Age <65	Older: Age ≥65	Younger: Age <65	Older: Age ≥65	Younger: Age <65	Older: Age ≥65
Incomplete Functional Recovery, <i>n</i> (%) ^a	160/258	26/76	146/240	20/69	131/225	14/66
Post-Concussive Symptoms ^b	132/259	11/74	120/239	14/66	107/221	11/59
Depressive Symptoms ^c	113/258	14/74	90/239	15/65	83/218	14/58

^aIncomplete functional recovery defined as Glasgow Outcome Scale Extended Score of < 8.

^bPost-concussive symptoms defined as endorsing at least two symptom categories on the Rivermead Post-Concussion Questionnaire.

^cDepressive symptoms defined a score of > 5 on the Patient Health Questionnaire-9.

Table 4. Mixed effects logistic regression model on dichotomized age effects.

Variable	Odds Ratio ^a	Odds Ratio ^b	Odds Ratio ^c	Odds Ratio ^d
GOSE	7.554 (3.088–18.479)	0.035 (0.006–0.206)	0.790 (0.675–0.925)	0.813 (0.592–1.116)
RPQ	0 (0.666–2.026)	0 (0.008–0.134)	0.003 (0.831–1.042)	0.2 (0.86–1.48)
PHQ-9	1.162 (0.598–0.658)	.032 (0.028–0.357)	0.931 (0.777–0.974)	1.128 (0.967–1.619)
	0.125	0	0.016	0.088

Values are Odds ratio (confidence interval) *p*-value.

^aOdds of having the outcome given young.

^bDifference in odds at baseline between old and young.

^cRate of change in odds ratio of having the outcome over time for younger.

^dDifference in rate of change in odds ratio between old and young.

probabilities rather than fitted probabilities derived from logistic regression. Proportions of older vs younger participants above and below cut-off levels for outcome variables are compared in Table 3.

The odds of incomplete functional recovery for younger (OR = 0.790; CI = 0.675–0.925; log odds slope = −0.236, *p* = 0.003) and older individuals (OR = 0.64 CI = 0.482, 0.855; log odds slope = −0.443, *p* = 0.002) decreased significantly over time. Odds of having depressive symptoms decreased significantly over time for younger (OR = 0.870; CI = 0.777–0.974; log odds slope = −0.139 *p* = 0.016) but not older persons (OR = 1.09; CI = 0.86–1.37; log odds slope = 0.085, *p* = 0.47). The odds of having post-concussive symptoms did not change significantly over time for either age group (younger OR = 0.931; CI = 0.831–1.042; log odds slope = −0.072, *p* = 0.212) (older OR = 1.05; CI = 0.82–1.34; log odds slope = 0.049, *p* = 0.7) (Table 4).

The difference in rate of change over time in odds of functional improvement and post-concussive/depressive symptom decline in older adults was not significantly different from that of younger persons (odds ratio for age*time interaction: OR = 0.813; CI = 0.592–1.116, *p* = 0.200; OR = 1.128; CI = 0.86–1.48, *p* = 0.385; OR = 1.251; CI = 0.967–1.619, *p* = 0.088) (Table 4). Figures 2–4 are visual

representations of outcome trajectory across all time points for both age groups.

Discussion

This study investigated age differences in TBI outcomes with a sample of 447 adults meeting VA/DoD criteria for mTBI. By utilizing a diagnostically specific sample, we aimed to highlight contrasts in depressive symptoms, post-concussive symptoms, and functional recovery over time in older (≥65 years of age) vs younger (<65 years) individuals. In accordance with the historical opinion that increased age acts as a risk factor after TBI, our hypothesis projected that, across time, older adults would have significantly worse outcomes than their younger counterparts.

Findings ran contrary to our hypothesis, with the rate of change over time in odds of improvement lacking statistical difference between age groups. Therefore, it can be said that the older group had effectively equivalent mood/post-concussive symptomatology and functional recovery to younger adults at all three successive time points post-head injury. Though the older adult group did not have significantly better improvement than younger individuals, this remains a provocative finding in the face of numerous publications supporting the viewpoint that increased age is predominantly detrimental to recovery following head injury (Czosnyka et al., 2005; Hukkelhoven et al., 2003; LeBlanc et al., 2006; Livingston et al., 2005; Mushkudiani et al., 2007; Perdue et al., 1998). Individually, each age group significantly improved in functional ability, but not in level of post-concussive symptomatology. Depressive symptoms significantly decreased over time for younger but not older adults.

Even within our exclusively mTBI sample, there likely exists a spectrum of intra-diagnostic severity. This range could originate from several possible sources. Older adults in this sample were disproportionately affected by falls, vs motor vehicle/traffic accidents being the most common mechanism of

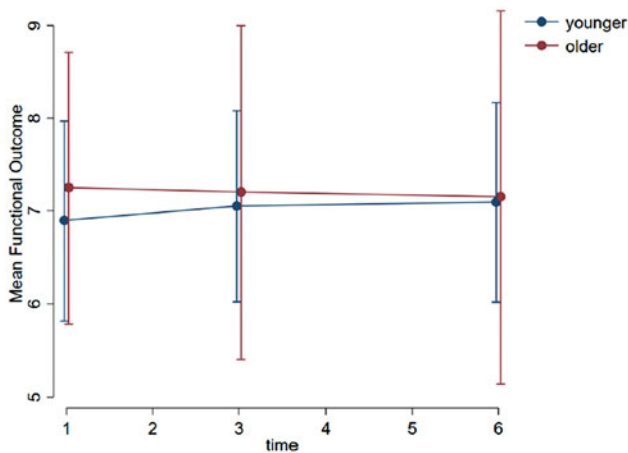


Figure 2. Functional recovery by time point.

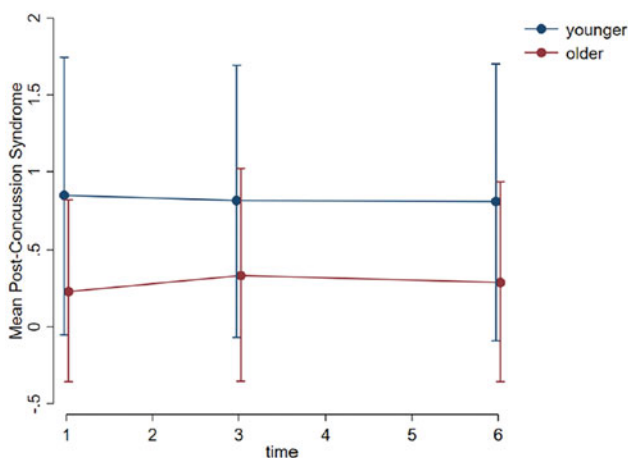


Figure 3. Post-concussive symptoms by time point.

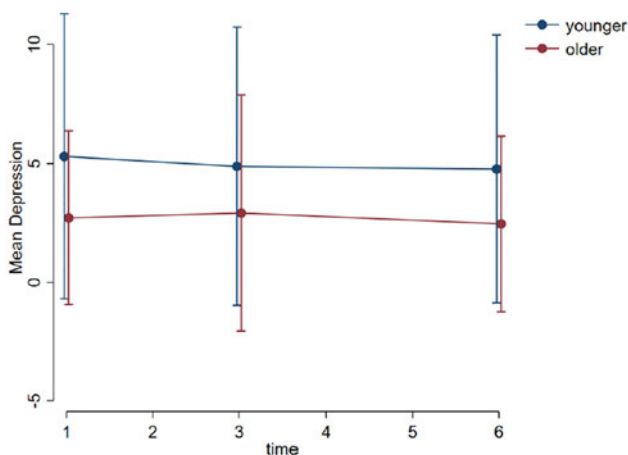


Figure 4. Depressive symptoms by time point.

injury for younger individuals. The potential that a higher velocity traffic accident could cause more brain damage than a fall from standing height could be acting as a confounder to results. It is also possible that older individuals with blunt head trauma are more likely to be brought to the ED than younger persons,

meaning that less symptomatic individuals in the younger age group who did not come to the ED were not part of our study. Both of these ideas circle around the notion that younger individuals in our sample may be comparatively more impaired rather than older persons being more resilient than expected. The potential of extant protective factors in older individuals is more difficult to justify, with one preliminary hypothesis pertaining to the brain to cerebrospinal fluid ratio being altered in aged samples with brain atrophy. It is possible that this could diminish the intensity of a blunt impact, however this idea has yet to be quantitatively investigated.

The design of this study is not without limitations. As is a common constraint of TBI research, this study does not have a true pre-injury baseline time point. This limits the ability to adjust observed outcome trajectories for pre-existing functional ability, depressive symptoms, and neurological symptoms that mimic those found in post-concussive presentations. Also in relation to outcomes, two of our measures, the GOSE and PHQ9, are not TBI-specific. They are also limited in that the GOSE has been criticized for its ceiling effects in less severely injured populations, and the PHQ9 does not allow us to comment on diagnostic criteria for depressive disorder. Regarding delineation between age groups, the cut-point of 65 years was chosen out of convention in the ageing literature, as opposed to a more optimal, TBI-centric justification. There is no current consensus on the 'critical threshold age' at which the risk profile following TBI changes, with studies suggesting anywhere from 35–75 years old (Dhandapani, Manju, Sharma, & Mahapatra, 2012). While our sample sizes are respectable, the older adult group only comprised ~20% of the overall sample. Additionally, we did not evaluate for cognition or medication use here, which may impact symptom reporting and differ between older and younger individuals. This study did not include data on pre-morbid medical problems, frequency and length of hospitalization following TBI, and other injuries associated with the TBI, which are possible confounding factors.

The results of this study contribute to the growing evidence base that a sub-set of older individuals can have outcomes after TBI that are comparable to or better than that of younger persons (Peters et al., 2018; Rapoport & Feinstein, 2001; Rapoport et al., 2003; Stapert et al., 2006). This is in accordance with our previous study, which found equivalent to superior outcomes at 1 month amongst older adults with any TBI severity by VA/DoD criteria (Peters et al.,

2018). Further studies are warranted to distinguish sub-groups of high-recovery older adults and the inherent mechanisms behind this phenomenon, such as protective factors or bias in diagnostic criteria. Through these efforts, and in the wake of the increasing utilization of healthcare services by older individuals, those requiring more aggressive treatment than the current standard can be identified.

Disclosure statement

The authors report no declarations of interest relevant to this manuscript. The data that support the findings of this study can be obtained from the senior author, MEP, upon reasonable request.

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